EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 03

Vectors

Magnitude and Dot Product

By - Excellence Programme

1 October 2026

Topics to be covered

01 Magnitude from components

02 Triangle law of addition

03 Resultant formula and range

04 Equal magnitudes

05 Subtraction of vectors

06 Addition by components

07 Dot product and angle

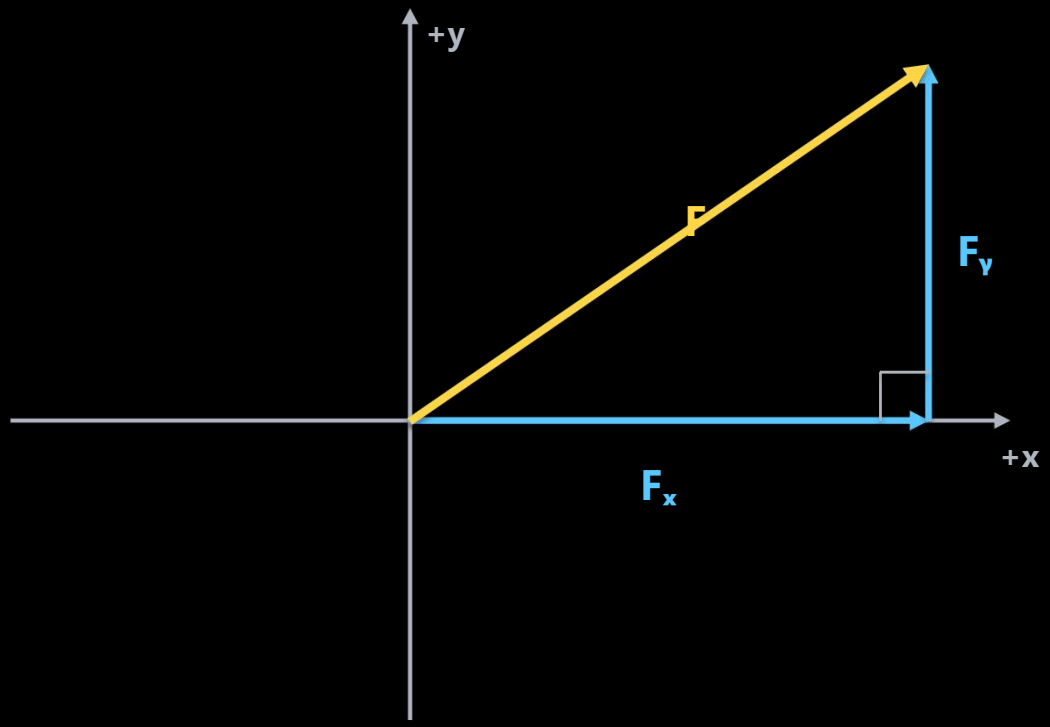
08 Unit vector

Magnitude from components

$$|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$$

Magnitude is the length of the diagonal

Components form a right triangle. The vector is the diagonal, so Pythagoras gives its length.



$$F^2 = F_x^2 + F_y^2 \quad \text{so} \quad |F| = \sqrt{(F_x^2 + F_y^2)}$$

With z-component

In 3D, add one more perpendicular square: $|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$.

Magnitude of 30 î + 40 ĵ

Find the magnitude of F = 30 î + 40 ĵ.

Solution: Magnitude of 30 î + 40 ĵ

Step 1: $|F| = \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)}$

Step 2: $\sqrt{2500} = 50$

50

Rough work / extra work

Magnitude when a component is negative

Find $|2\hat{\imath} + 2\hat{\jmath}|$, $|10\hat{\imath} + 10\hat{\jmath}|$ and $|3\hat{\imath} - 4\hat{\jmath}|$.

Solution: Magnitude when a component is negative

Step 1: $\sqrt{(4 + 4)} = \sqrt{8} = 2\sqrt{2}$

Step 2: $\sqrt{(100 + 100)} = 10\sqrt{2}$

Step 3: $\sqrt{(9 + 16)} = \sqrt{25} = 5$. The minus does not enter the square.

$2\sqrt{2}$, $10\sqrt{2}$, and 5.

Three components

Find $|3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}|$ and $|6\hat{\imath} - 8\hat{\jmath} + 10\hat{k}|$.

Solution: Three components

Step 1: $\sqrt{(9 + 16 + 25)} = \sqrt{50} = 5\sqrt{2}$

Step 2: $\sqrt{(36 + 64 + 100)} = \sqrt{200} = 10\sqrt{2}$

$5\sqrt{2}$ and $10\sqrt{2}$.

Rough work / extra work

Triangle law of addition

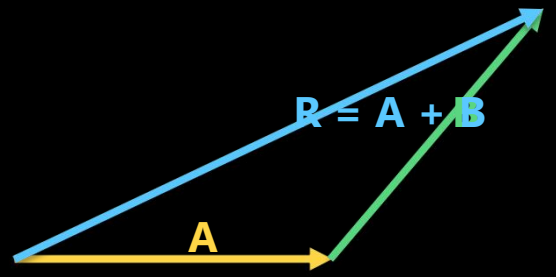
To find an angle, put tails together, or heads together.

To add, put the tail of the second arrow on the head of the first.

Resultant R runs from the first tail to the last head.

A + B = B + A. For three arrows the drawing order does not change R.

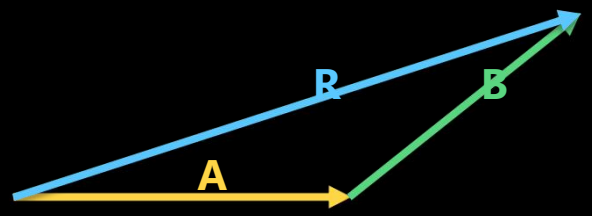
The resultant of two arrows lies between them, closer to the longer one.



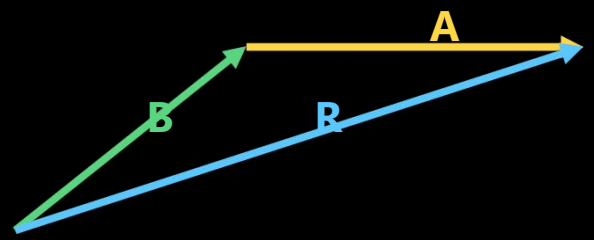
head-to-tail

Why drawing order does not change R

A + B and B + A reach the same final point. The path changes, but the start and finish do not.



A then B

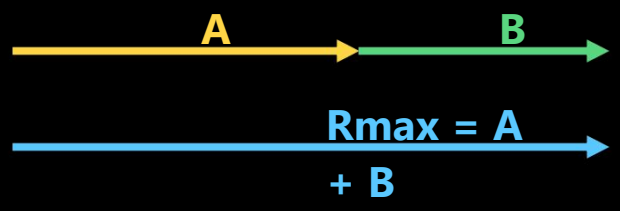


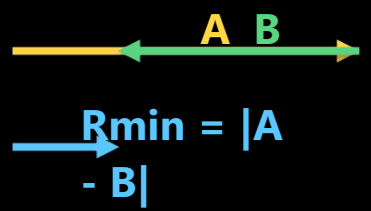
B then A

Same starting point + same ending point = same resultant.

Resultant range: minimum to maximum

The largest resultant happens when both arrows pull in the same direction. The smallest happens when they oppose.





Allowed range

$|A - B| \le R \le A + B$

A resultant outside this range is impossible.

Magnitude of the resultant

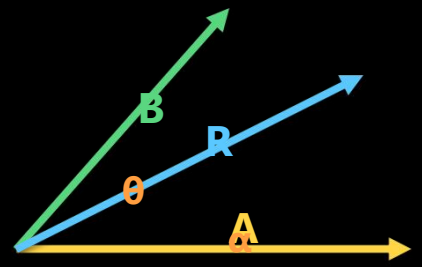
$$R = \sqrt{(A^2 + B^2 + 2\,A\,B\cos\theta)}$$

$$\tan\alpha = (B\sin\theta) / (A + B\cos\theta)$$

α is the angle between A and the resultant.

θ = 0: R max = A + B. θ = 180: R min = |A - B|. θ = 90: R = $\sqrt{(A^2 + B^2)}$.

R is allowed only from |A - B| to A + B. Both ends are included.



Two 4 N forces at 60°

|A| = 4 N, |B| = 4 N, angle = 60°. Find R.

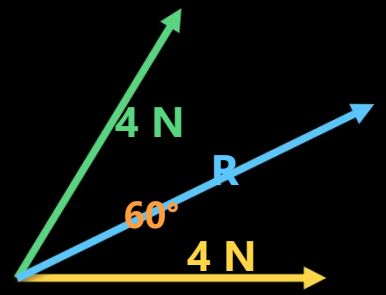
Solution: Two 4 N forces at 60°

Step 1: $R = \sqrt{(16 + 16 + 32 \times \cos 60°)}$

Step 2: $\cos 60° = 1/2$, so the middle term is 16

Step 3: $R = \sqrt{48} = 4\sqrt{3}$

$R = 4\sqrt{3}$ N.



Rough work / extra work

Magnitudes 3 and 4

Two vectors have magnitudes 3 and 4. Find R at 0°, 90° and 180°.

Solution: Magnitudes 3 and 4

Step 1: At 0°, R = 3 + 4 = 7

Step 2: At 90°, R = √(9 + 16) = 5

Step 3: At 180°, R = 4 - 3 = 1

7, 5 and 1.

3 and 4 at 60° and 120°

Same magnitudes 3 and 4. Find R at 60° and at 120°.

Solution: 3 and 4 at 60° and 120°

Step 1: At 60°: √(9 + 16 + 12) = √37

Step 2: At 120° cos = -1/2, so √(9 + 16 - 12) = √13

√37 and √13.

Rough work / extra work

5 N east and 10 N north

Two forces, 5 N due east and 10 N due north. Find the resultant.

Solution: 5 N east and 10 N north

Step 1: The angle is 90°, so the cross term is 0

Step 2: $R = \sqrt{(25 + 100)} = \sqrt{125} = \sqrt{(25 \times 5)} = 5\sqrt{5}$

$5\sqrt{5}$ N.

Two 5 N forces at 120°

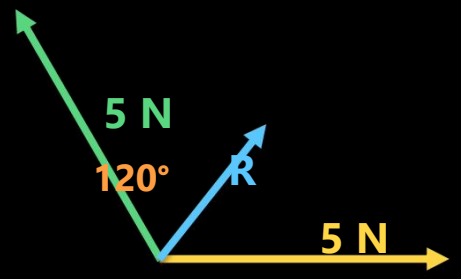
Two forces of 5 N have 120° between them. Find the net force.

Solution: Two 5 N forces at 120°

Step 1: Use the angle between the arrows, which is 120°, not a nearby 60° mark

Step 2: $R = \sqrt{(25 + 25 + 50 \times (-1/2))} = \sqrt{(50 - 25)} = \sqrt{25}$

Net force = 5 N.



Which pair cannot give a resultant of 4?

Pairs: (2, 4), (4, 4), (4, 9), (1, 5). Which cannot give resultant 4?

Solution: Which pair cannot give a resultant of 4?

Step 1: (2, 4) allows 2 to 6. 4 is inside.

Step 2: (4, 4) allows 0 to 8. (1, 5) allows 4 to 6.

Step 3: (4, 9) allows 5 to 13. 4 is smaller than the minimum.

(4, 9) cannot give 4.

Rough work / extra work

Three forces, zero resultant

Which set can have zero resultant: (2, 4, 7), (3, 1, 5), (2, 8, 11), or (3, 4, 2)?

Solution: Three forces, zero resultant

Step 1: The third force must lie between |first - second| and first + second

Step 2: For 3 and 4, the range is 1 to 7. The force 2 lies inside it.

Step 3: 7 is outside 2 to 6. 5 is outside 2 to 4. 11 is outside 6 to 10.

Only (3, 4, 2) can give zero.

Equal magnitudes

If |A| = |B| = A, then R = 2 A cos(θ / 2)

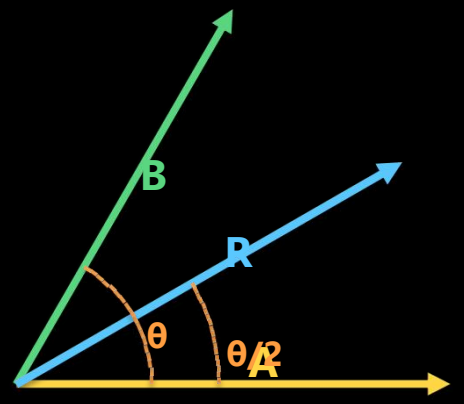
Their difference has magnitude D = 2 A sin(θ / 2)

The resultant then bisects the angle. Angle between A and R is θ / 2.

θ = 0: R = 2A, D = 0. θ = 90: both are √2 A. θ = 120: R = A and D = √3 A. θ = 180: R = 0 and D = 2A.

Equal vectors: resultant bisects the angle

When |A| = |B|, the triangle is symmetric. The resultant must lie exactly in the middle.



$$R = 2A \cos(\theta / 2)$$

Fast memory

Equal vectors make an isosceles triangle, so R is the angle bisector.

Rough work / extra work

Sum and difference have the same magnitude

If |A + B| = |A - B|, what is the angle between A and B?

Solution: Sum and difference have the same magnitude

Step 1: Equal lengths of sum and difference means the cross term must vanish

Step 2: $2AB\cos\theta = 0$, so $\cos\theta = 0$

$90°$.

Two unit vectors add to a unit vector

The sum of two unit vectors is itself a unit vector. Find the magnitude of their difference.

Solution: Two unit vectors add to a unit vector

Step 1: $R = 1$ and each magnitude is 1, so $1 = 2\cos(\theta / 2)$, so $\theta = 120°$

Step 2: $D = 2 \sin 60° = 2 \times \sqrt{3}/2 = \sqrt{3}$

Magnitude of the difference = $\sqrt{3}$.

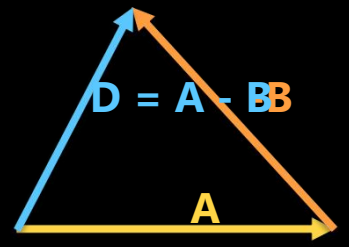
Vector subtraction

A - B = A + (-B). Reverse B, then add.

$$D = \sqrt{(A^2 + B^2 - 2\,A\,B\cos\theta)}$$

$\theta = 0$: $D = |A - B|$, the minimum. $\theta = 180$: $D = A + B$, the maximum. $\theta = 90$: $D = \sqrt{(A^2 + B^2)}$.

A + B equals B + A. A - B does not equal B - A.

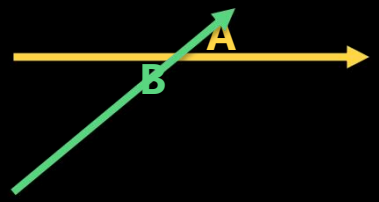


reverse B first

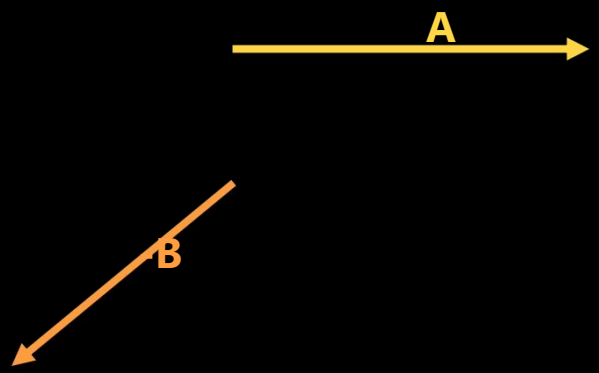
Rough work / extra work

Subtraction is not a new operation

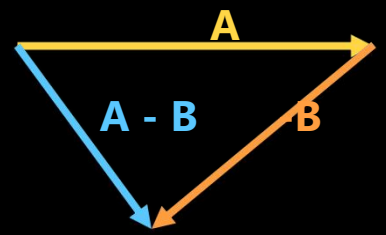
Every subtraction question is an addition question after reversing the second vector.



Step 1: given A and B



Step 2: reverse B

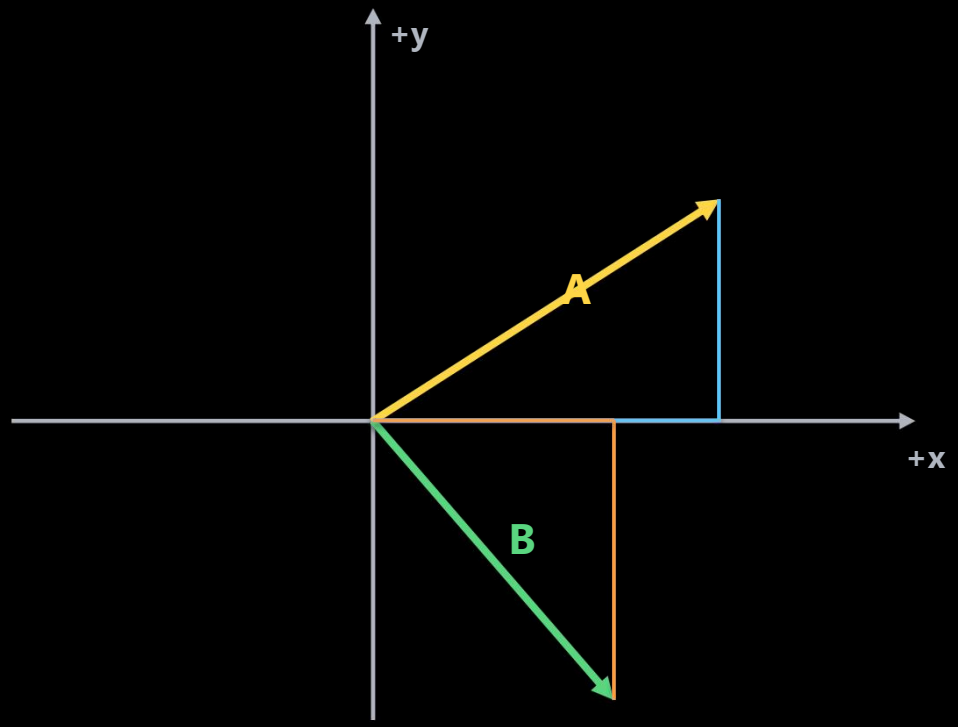


Step 3: add A + (-B)

A - B and B - A have the same length but opposite directions.

Adding by components is adding columns

The x-parts combine with x-parts. The y-parts combine with y-parts. This is the cleanest method for many vectors.



$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\,\hat{\imath} + (A_y + B_y)\,\hat{\jmath}$$

Column idea

x-column decides left-right. y-column decides up-down.

3D same rule

$A + B = (A_x+B_x)\hat{\imath} + (A_y+B_y)\hat{\jmath} + (A_z+B_z)\hat{k}$

Add by components

$A = 4\hat{i} + 3\hat{j}$ and $B = 3\hat{i} - 4\hat{j}$. Find $A + B$, $A - B$, and $|A| + |B|$.

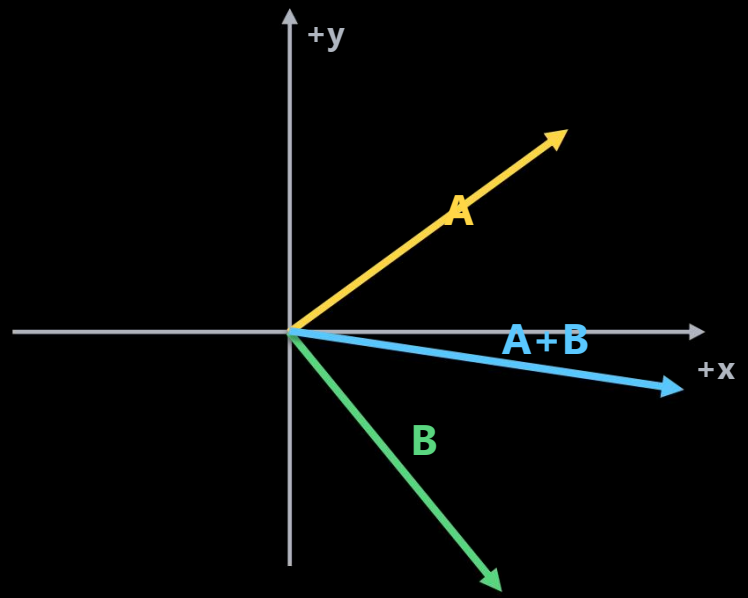
Solution: Add by components

Step 1: $\vec{A} + \vec{B} = 7\,\hat{i} - 1\,\hat{j}$

Step 2: $\vec{A} - \vec{B} = 1\,\hat{i} + 7\,\hat{j}$

Step 3: $|\vec{A}| = 5$ and $|\vec{B}| = 5$, so $|\vec{A}| + |\vec{B}| = 10$

Sum of magnitudes is 10. Magnitude of the sum is √50.



Change in velocity

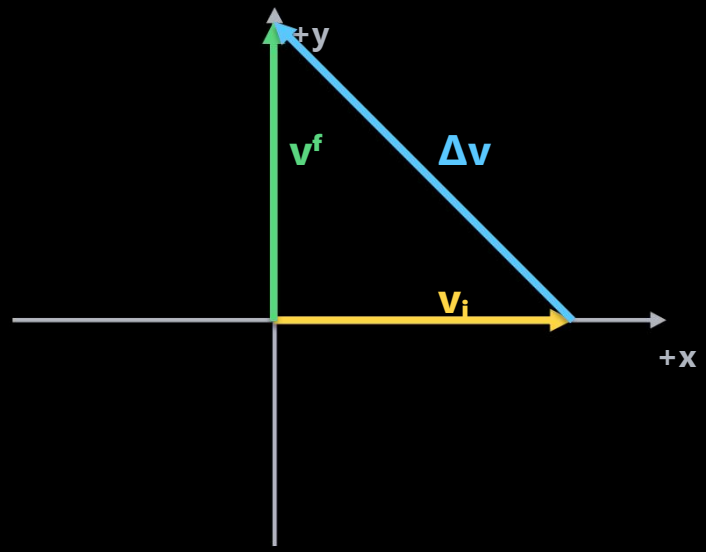
Velocity changes from 10 m/s east to 10 m/s north. Find the change in velocity.

Solution: Change in velocity

Step 1: Change = final - initial = 10 ĵ - 10 î

Step 2: Magnitude = √(100 + 100) = 10 √2

10 √2 m/s, pointing north of west.



Rough work / extra work

Rotate 2 î

A = 2 î is rotated by 90°, 180°, 270° and 360° anticlockwise. Write each new vector.

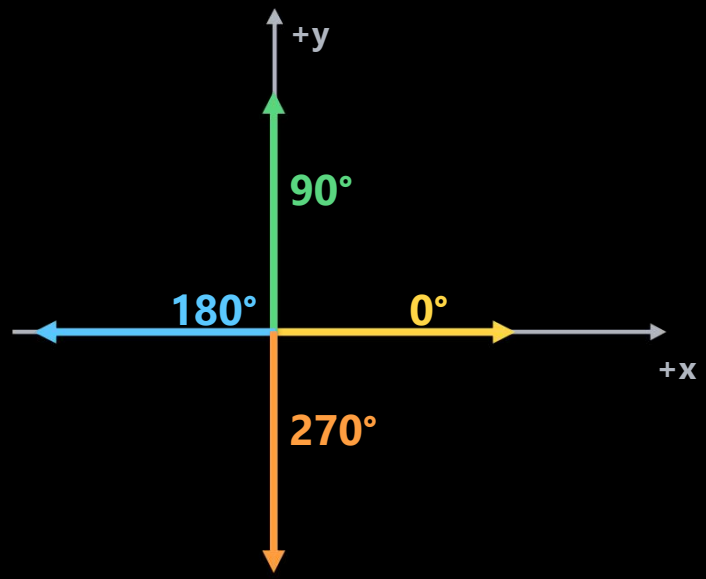
Solution: Rotate 2 î

Step 1: 90°: 2 ĵ

Step 2: 180°: -2 î

Step 3: 270°: -2 ĵ. 360°: 2 î again

Rotation keeps the magnitude and changes only the direction.



Dot product

$$\mathbf{A} \cdot \mathbf{B} = A\,B\cos\theta = A_x B_x + A_y B_y + A_z\,B_z$$

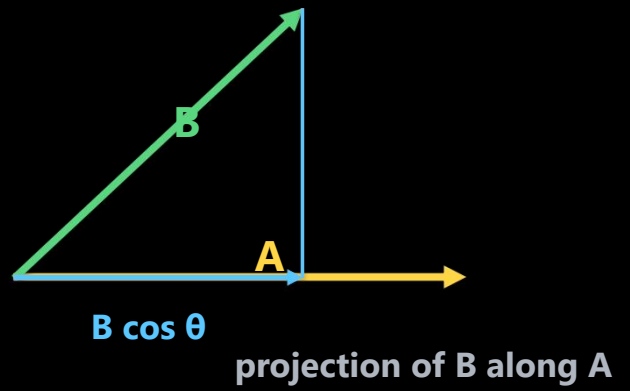
If the dot product is 0, the vectors are perpendicular.

cos θ = (A · B) / (A B). That is how the angle is found from components.

Component of B along A = (A · B) / A.

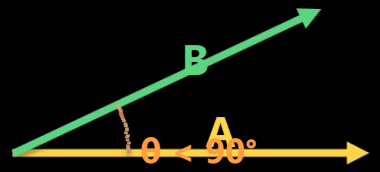
$\hat{i} \cdot \hat{i} = 1$. $\hat{i} \cdot \hat{j} = 0$. The same rule holds for $\hat{j}$ and $\hat{k}$.

A unit vector dotted with itself is 1.

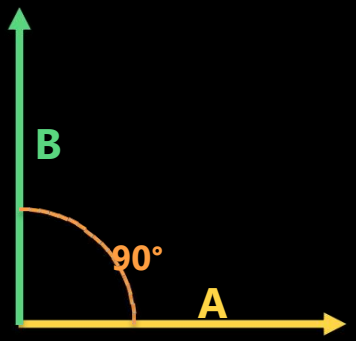


Dot product tells alignment

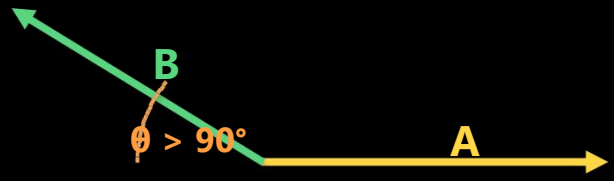
Dot product is positive when vectors face mostly the same way, zero at 90°, and negative when they face opposite way.



$A \cdot B > 0$



$A \cdot B = 0$



$A \cdot B < 0$

Why this matters

It quickly tests perpendicular vectors and helps find the angle between two vectors.

Rough work / extra work

Are these two vectors perpendicular?

A = 3 î + 4 ĵ + 5 k̂ and B = 3 î + 4 ĵ - 5 k̂. What is the angle between them?

Solution: Are these two vectors perpendicular?

Step 1: $A \cdot B = 9 + 16 - 25 = 0$

The angle is 90°.

Angle with 4 $\hat{\imath}$

$A = 2\hat{\imath} + 6\hat{\jmath} + 3\hat{k}$ and $B = 4\hat{\imath}$. Find $\cos\theta$.

Solution: Angle with 4 î

Step 1: $A \cdot B = 8$

Step 2: $|A| = \sqrt{(4 + 36 + 9)} = 7$ and $|B| = 4$

Step 3: $\cos \theta = 8 / 28 = 2/7$

$\theta = \cos^{-1}$ of 2/7.

Unit vectors from components

$\hat{i}$ itself is already a unit vector.

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$.

Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$.

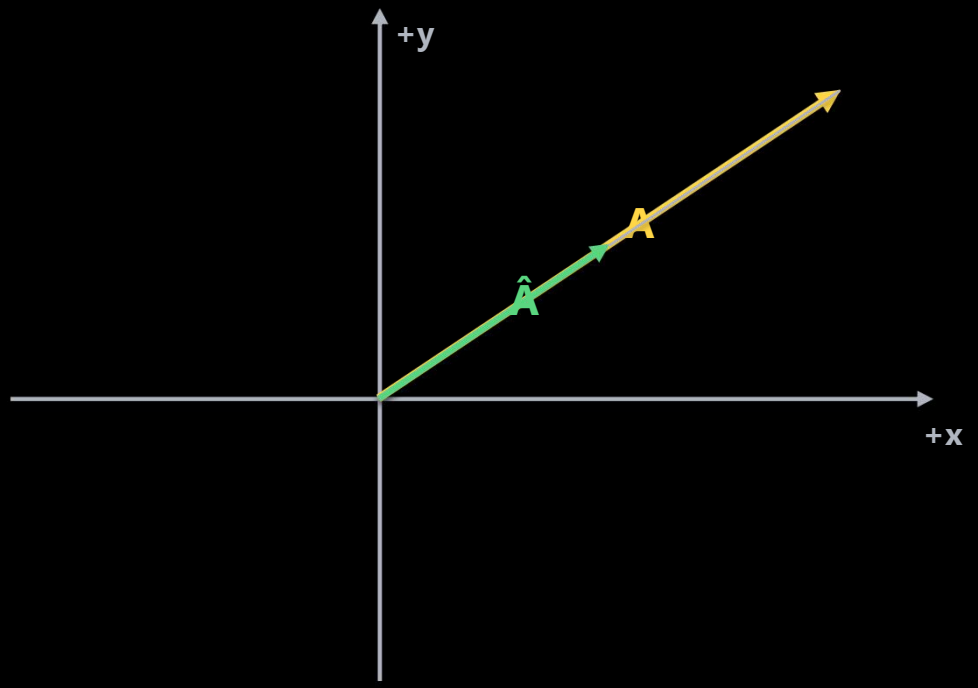
$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1, because $\sin^2 + \cos^2 = 1$.

Unit vector of $3\,\hat{i} + 4\,\hat{j}$ is $(3\,\hat{i} + 4\,\hat{j}) / 5$.

Rough work / extra work

Unit vector keeps direction and removes size

A unit vector is only the direction of a vector. It has length 1 and no physical unit.



$$\hat{A} = A / |A|$$

Example

For $A = 3\hat{i} + 4\hat{j}$, $|A| = 5$, so $\hat{A} = (3\hat{i} + 4\hat{j})/5$.

Do not attach N, m/s or kg to a unit vector.

Perpendicular forces A and A/2

Two forces of magnitudes A and A/2 are perpendicular. Find the magnitude of their resultant.

Solution: Perpendicular forces A and A/2

Step 1: $\cos 90° = 0$, so $R = \sqrt{(A^2 + A^2/4)}$

Step 2: $R = \sqrt{(5\,A^2 / 4)} = (A/2)\,\sqrt{5}$

$(\sqrt{5} / 2) \times A$.

Resultant is half the larger vector

The resultant of two vectors is half the larger one, and it is perpendicular to the smaller one. Find the angle between the vectors.

Solution: Resultant is half the larger vector

Step 1: R perpendicular to the smaller vector means $(A + B) \cdot B = 0$

Step 2: $\cos\theta = -B/A$

Step 3: $R = A/2$ then gives $B = (\sqrt{3}/2)\,A$

$\cos\theta = -\sqrt{3}/2$, so $\theta = 150°$.

Rough work / extra work

Acceleration of a 4 kg body

$F_1 = 5\,\hat{i} + 8\,\hat{j} + 7\,\hat{k}$ and $F_2 = 3\,\hat{i} - 4\,\hat{j} - 3\,\hat{k}$ act on 4 kg. Find the acceleration.

Solution: Acceleration of a 4 kg body

Step 1: Net force = 8 $\hat{i}$ + 4 $\hat{j}$ + 4 $\hat{k}$

Step 2: a = net force / 4

a = 2 $\hat{i}$ + 1 $\hat{j}$ + 1 $\hat{k}$.

Vector 03 lock lines

Components: cos along the axis you measured from, then keep the sign

$|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$

$R = \sqrt{(A^2 + B^2 + 2AB \cos \theta)}$, and $|A-B|$ to $A+B$ is the allowed range

$A \cdot B = 0$ means 90°. Unit vector = vector divided by magnitude.

Practice

1. Find $|3\,\hat{i} + 4\,\hat{j} + 5\,\hat{k}|$ and $|6\,\hat{i} - 8\,\hat{j} + 10\,\hat{k}|$.

2. Redo magnitudes 3 and 4 at 0°, 60°, 90°, 120° and 180°.

3. Redo the unit-vector list and the 4 kg acceleration.

4. If $|A + B|$ equals $|A - B|$, write the angle between A and B.

Do not add magnitudes when the arrows are not along one line.